And the winner is...

In the end, we found that although the Princeton drives topped all our tests, their absolute lack of bundled accessories—not even a driver CD—ensured a low overall score. If you are looking for a drive with sheer performance, and software and accessories really don't matter, opt for the Princeton MPDUSB64MB drive.

On the other hand, the 128 MB Easy Disk drive gave an above-average showing in the performance tests, and with its extra software, won the Best Performance award. The Transcend 512 MB drive chalked up some decent scores in our tests, and has a low cost per MB of just Rs 18.22. Thus, it was the clear winner for the Best Value award.

EXTERNAL HARD DRIVES

Purely in terms of mode of operation and drive mechanics, there is virtually no difference between internal and external hard disks. However, with a USB or FireWire interface, an external hard drive packs in greater flexibility and ease of installation. For our comparison, we received eight drives, out of which four were of the faster 7,200-rpm type, and the rest were smaller drives with speeds of 4,200 rpm.

Features

The one thing that stood out in the hard drive comparison was that all the drives we tested were USB 2.0 compliant—unlike the USB drive comparison. The Iomega 120GB also supported a FireWire interface, and was the largest capacity drive we got. It was heavy and bulky, but the heaviest were the 40 GB and 80 GB Freecom FHD-1 models, weighing 900 gms. The heavy Iomega and Freecom drives also depended on an external power source, confirming that they are more suited for desktop use. At the other end of the scales, were the two Freecom FHD-2 models, weighing 190 gms.

The Freecom FHD-1 40 GB offered simple data synchronisation and an excellent overall performance to bag the Best Performance award

Despite being the lightest in the test group, the FHD-2 drives still looked far sturdier than their desktop equivalents, especially with their smooth and sleek bodies. The FHD-1 drives, in comparison, only managed to look good with evenly contoured bodies and cool blue LEDs on the front panel. The Transcend drives were also very light, and were the only ones to give directions on how to replace the internal disks from within their casing. All the 4,200-rpm hard drives had a smaller 2.5-inch form factor, as they use mobile hard drives. Hopefully, the recent release of the faster 7,200-rpm mobile drives, with the 2.5-inch form factor, will improve the external drive products.

In terms of added functionality, the Freecom FHD-1 drives clearly had the upper hand with a data Sync button on the front panel. This button allows synchronisation of data between the external device and the system disk, with the help of the bundled FSync 2.0 software. FSync is fully compatible with Microsoft Briefcase, and can be used to easily transfer your data and synchronise your files between computers. As for the software bundled, the Freecom FHD-2 drives came packaged with Acronis TrueImage, a utility for creating images and snapshots of hard drives on external drives. The drives from Iomega came with a lot more, including Iomega's own Automatic Backup software as well as Norton Ghost 2003. Another useful utility is the Iomega FAT32 formatter, which can format large drives and partitions.

The Transcend drives were disappointing, and only provided a driver CD, power and USB cables. However, they came with something that no other drive—portable or external—could match: they were packed in a stylish leather case, along with the requisite data cable and also a visiting card-sized driver CD. The inclusion of the carry-case clearly added a lot of value to the Transcend drives, increasing their portability factor.

Performance

Let's take a look at how the drives did in each of the performance tests we put them through:

SiSoft Sandra Professional 2003: In the Sisoft benchmark

USB 2.0 v/s FireWire

Out of the eight external hard drives we received, the Iomega 120 GB model was the only drive which featured a FireWire interface in addition to USB 2.0. The difference in the theoretical transfer rates between the IEEE 1394 standard and the USB 2.0 interface is only 80 MBps, with the latter running at 480 MBps. To ascertain the effects of this difference in an actual test scenario, we ran our tests again on the Iomega drive, but this time, we connected it through the FireWire interface.

Surprisingly, the Sisoft Sandra scores put USB 2.0 on top, with a drive index of 22,308 KBps as compared to 18,703 KBps with FireWire. The USB interface won again in the sequential read and write speed tests with scores of 32 MBps and 26 MBps respectively, as compared to the slower 27 MBps and 21 MBps with FireWire.

In the HDtach benchmarks, FireWire scored better with an average write speed of 20.5 MBps, as opposed to 17.3 MBps with USB 2.0. Also, HDTach reported a 0 per cent CPU utilisation with FireWire, as compared to 18.3 per cent using USB.

In the real-world tests, the USB interface was the clear winner as FireWire was barely able to have a faster assorted write speed taking 58.76 seconds to write a 1 GB folder, as compared to 59 seconds in our primary tests. The real difference in the two standards was highlighted in the sequential write test: with the FireWire interface it took seven seconds longer to copy our 1 GB test image than with USB—103 seconds with FireWire and 96.6 seconds with USB 2.0.